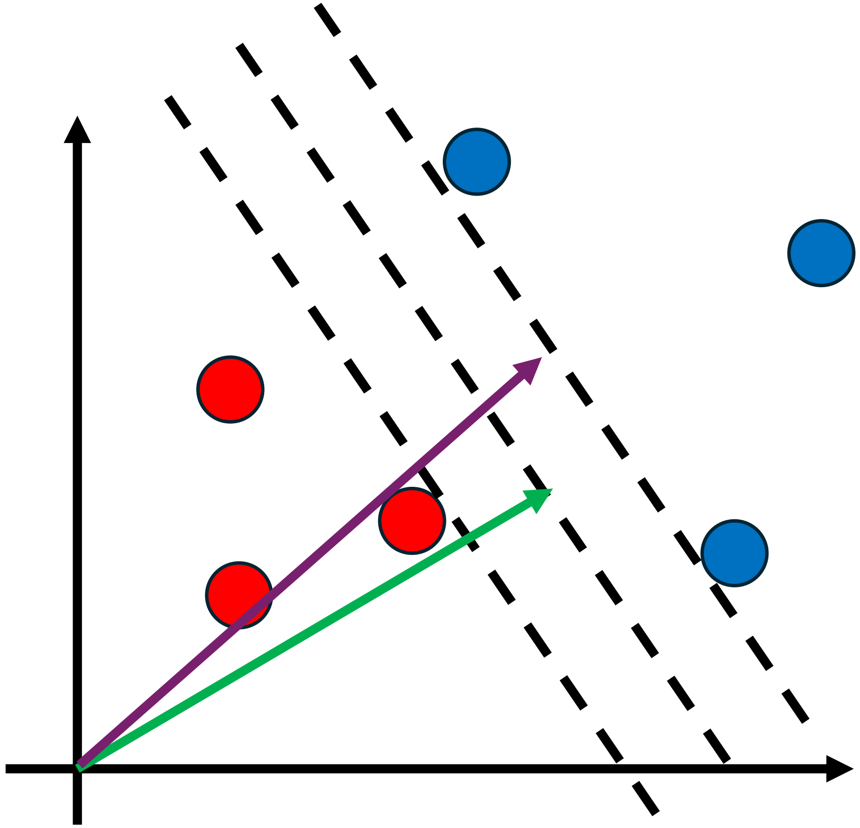
Support Vector Machines

Machine Learning

ASSUMPTION: data are linearly separable



IDEA: The best boundary is the one that gives the highest margin between the two classes

$$w^T x + b \geq 0$$ classification rule

$$y_i \in \{-1,1\}$$

sample with y_i =+1

$$w^T x_+ + b \geq \gamma$$

to impose a margin

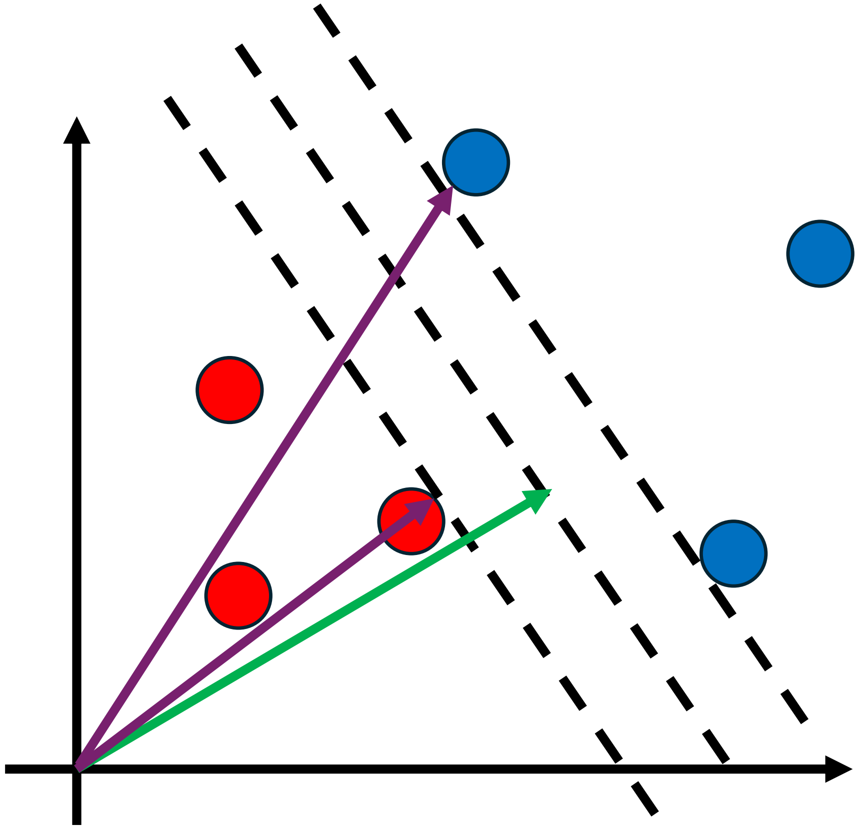
$$w^T x_- + b \leq -\gamma$$

sample with y_i =-1

$$y_i(w^T x_i + b) - \gamma \geq 0$$

$$y_i(w^T x_i + b) - \gamma = 0$$ for the samples at the boundaries

We want to find w and b that maximize the width of the margin



sample at the boundary of the region y_i =+1

$$(x_+ - x_+) \cdot \frac{w}{|w|}$$

This is the width of the margin

sample at the boundary of the region y_i =-1

$$(w^T x_+ + b) - \gamma = 0$$

$$w^T x_+ = \gamma - b$$

$$-(w^T x_- + b) - \gamma = 0$$

$$-w^T x_- = \gamma + b$$

$$(x_+ - x_+) \cdot \frac{w}{|w|} = \frac{2\gamma}{|w|}$$

- Gamma is a multiplicative factor, we can assume gamma = 1 without loss of generality
- So to make the margin wide we need to minimize the norm of w
- It’s the same as minimizing 0.5*|w|^2

$$y_i(w^T x_i + b) - 1 \geq 0$$

constraint to satisfy

$$f = \frac{1}{2}|w|^2$$

function to minimize

it's the same as minimizing this function using Lagrange multipliers

$$L = \frac{1}{2}|w|^2 - \sum \alpha_i [y_i(w^T x_i + b) - 1]$$

This sum is over all the points, but the constraint is strictly positive for points not in the margin, and so in this case alpha needs to be zero, while it needs to be strictly positive for samples at the boundary

setting the derivatives to zero

$$\frac{\partial L}{\partial w} = w - \sum \alpha_i y_i x_i \longrightarrow w = \sum \alpha_i y_i x_i$$

$$\frac{\partial L}{\partial b} = -\sum \alpha_i y_i \longrightarrow \sum \alpha_i y_i = 0$$

$$L = \frac{1}{2}\left(\sum \alpha_i y_i x_i\right)\left(\sum \alpha_j y_j x_j\right) - \sum \alpha_i \left[y_i \left(\left(\sum \alpha_i y_i x_i\right) x_i + b\right) - 1\right]$$

$$L = \frac{1}{2}\left(\sum \alpha_i y_i x_i\right)\left(\sum \alpha_j y_j x_j\right) - \sum \alpha_i \left[y_i \left(\left(\sum \alpha_i y_i x_i\right) x_i + b \right) - 1 \right]$$

$$L = \frac{1}{2}\left(\sum \alpha_i y_i x_i\right)\left(\sum \alpha_j y_j x_j\right) - \sum \alpha_i y_i x_i \left(\sum \alpha_i y_i x_i\right) - \sum \alpha_i y_i \, b + \sum \alpha_i$$

it's zero

$$L = \sum \alpha_i - \frac{1}{2}\left(\sum \alpha_i y_i x_i\right)\left(\sum \alpha_j y_j x_j\right) = \sum \alpha_i - \frac{1}{2}\sum \alpha_i \alpha_j \, y_i y_j x_i \cdot x_j$$

The function that we want to minimize depends only on the scalar products of the samples

$$w^T x + b \geq 0$$

$$w = \sum \alpha_i y_i x_i \qquad \sum \alpha_i y_i x_i \cdot x_j + b \geq 0$$

The classification rule also depends only on the scalar product of the samples

The Kernel ”trick”

The function to minimize and the function used to make predictions depends on the samples only through their scalar product

- $x \longrightarrow \phi(x)$ Assume that a mapping exist from the original space to a new dimensional space (usually high dimensional)
- $\phi^T(x)\phi(z) = K(x, z)$ The function K gives the scalar product in the high dimensional space

Since the algorithm depends only on scalar product, if we know K we don’t need to work in the high dimensional space

How do we know that a function is a Kernel ?

In order for a function to be a Kernel, it needs to exist a mapping function $x \longrightarrow \phi(x)$ such that $K(x,z) = \phi(x)^T \phi(z)$

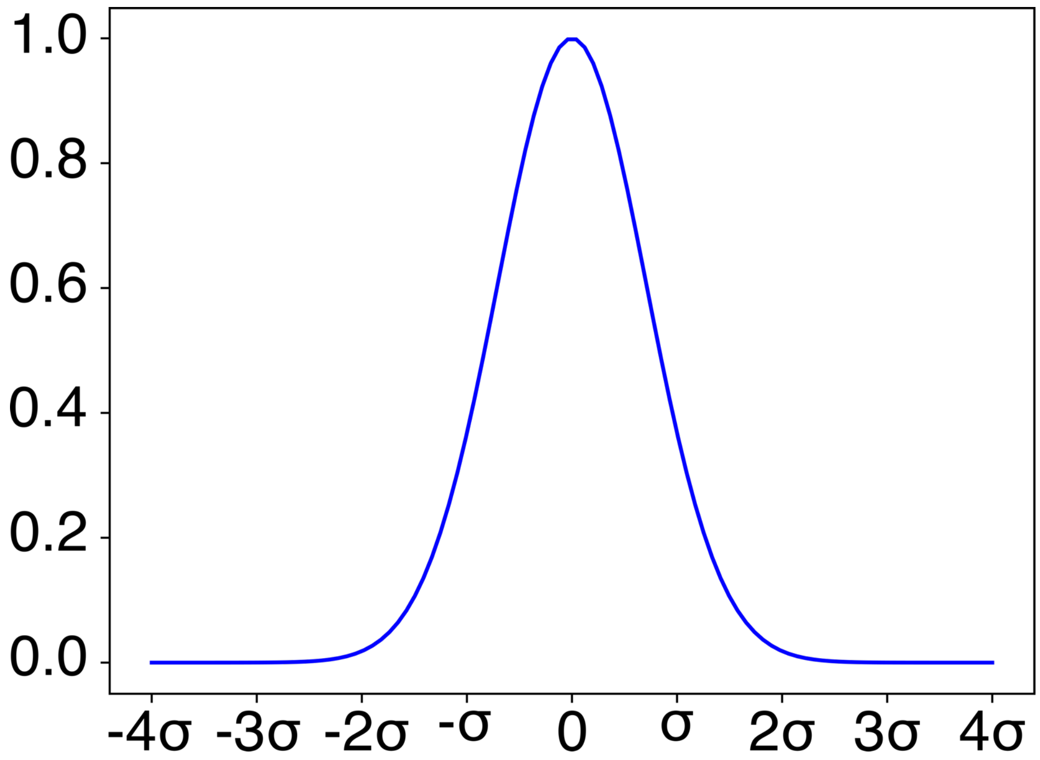
Mercer's theorem. A function is a Kernel if and only if given any set of points $x_1, \dots, x_d$ the matrix with elements $K_{ij} = K(x_i, x_j)$ is positive semi-definitive

Radial Basis Function kernel (Gaussian Kernel)

$$K(x, z) = e^{-\frac{\|x-z\|^2}{2\sigma^2}}$$

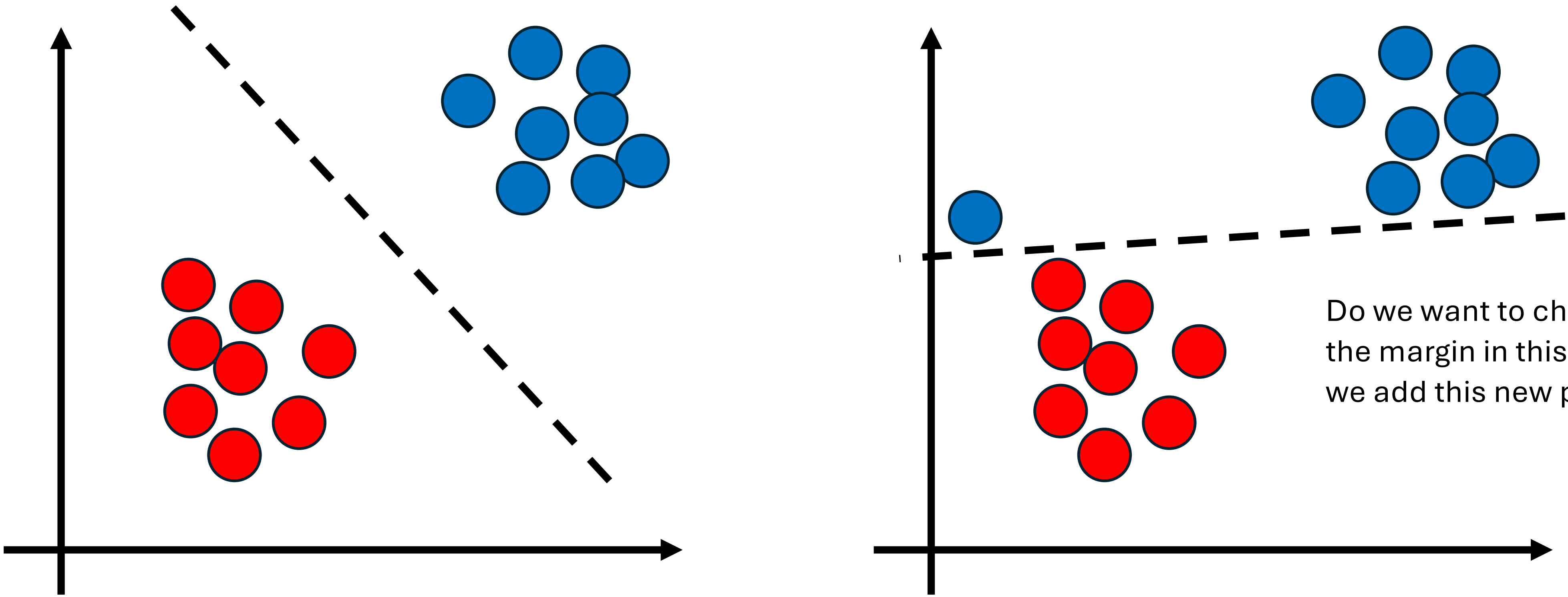
Intuition: It measures the similarity between two samples

It maps to an infinite dimension dataspace



Soft Margin Classifier

Perfect separation among all points is not always optimal



Do we want to change the margin in this way if we add this new point ?

Soft Margin Classifier

$y_i(w^T x_i + b) - 1 \geq 0$ constraint to satisfy

$f = \frac{1}{2}|w|^2$ function to minimize

$y_i(w^T x_i + b) - 1 \geq -\xi_i, \xi_i > 0$ some points now can be inside the margin

$f = \frac{1}{2}|w|^2 + C \sum \xi_i$ but we want to minimize the number of points inside the margin

high C → points inside the margin gives high penalty → low regularization
low C → points inside the margin gives low penalty → high regularization

Kernel PCA

The Kernel trick can be used in any algorithm that depends on input features only through the scalar products between samples, so for instance in PCA

- Compute the Kernel for matrix
- Centre the Kernel matrix (subtract means over columns and rows; then add back the global mean). This is needed because PCA assumes assumes the data are centred but.
- Compute eigenvalues/eigenvectors
- Project data onto the principal components (these are linear combinations of the features in the mapped space)

It can reduce dimensions even in the presence of non-linear relation

The principal components are not interpretable